

AI Food Recommendation Agent Assignment

Assignment – Designing Intelligent Agent Using PEAS framework

Description:

For this assignment, I have created an artificial intelligence agent that recommends a restaurant based on the user's location, budget, and cuisine to a user. The user provides the input, and the agent accesses the database of the restaurant and provides the matching results based on the user's input.

Problem Statement

It may be quite time-consuming to choose the best restaurant based on the location, budget, and cuisine. Sometimes it may also be challenging to balance between the given parameters for choosing the best restaurant. Thus, I have developed an agent that will help the user in finding the best matching restaurant based on their preference and budget.

PEAS Framework Analysis

PEAS stands for Performance measure, Environment, Actuators, and Sensors. It is used to provide a background description of the agent.

i: P: The performance of the agent can be defined based on the accuracy of the recommendation, fulfilling the user's budget and cuisine, the number of results retrieved, and the quick response of the agent.

ii: E: The environment of the agent in this problem includes the database of the restaurant, the user, the city, location, cuisine, and budget.

iii: A: The actuators of the agent include the display or printed output of the recommended restaurant, the address, budget category, and cuisine.

iv: S: The sensors of the agent include the input provided by the user in terms of location, budget, and cuisine.

Environment Properties:

Observable – Partially observable: The environment is partially observable because the agent can only access the database, and it does not have information about the current availability of the restaurant.

Deterministic: The environment is deterministic because if the user inputs the same query, then the result that the agent will provide will be the same because the database remains constant.

Static/Dynamic: The agent is static because the database does not change until the request is processed.

Single/Multi-agent: The environment is considered single agent because only the user and the AI agent are present.

Evaluation

Performance: I have tested the performance of the AI agent by providing various inputs of location, budget, and cuisine. The agent correctly retrieved the data, and if there was no matching record in the database, the agent responded with 'No record found. Thus, based on the test runs, I can say that the agent performs well on the given input and provides accurate results quickly.